General Sir John Kotelawala Defense University
Faculty of Computing
Department of Computer Engineering

BSc (Hons) in Computer Engineering
Laboratory Practical

Arduino Experiment 01 – Blinking an LED with Arduino

Name: ……

Index No ……

Intake ……

Date ……

Instructor Name and Signature: ……

Comments ……..
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Grade

Objectives:
To understand the basic operation of the Arduino microcontroller by writing and uploading a simple program to control an LED using a digital output pin.

Outcomes:
After completing this experiment, students would be able to:

- Identify basic Arduino hardware components.
- Write a simple Arduino program using the Arduino IDE.
- Upload a program to the Arduino board.
- Control an external electronic component (LED) using digital output.

Equipment Required:

Arduino Board, Arduino USB Cable, Breadboard, LED, 220 Ω Resistor, Jumper Wires

Theory

Arduino is an open-source microcontroller platform used for building electronic projects. It consists of a programmable circuit board and software called the **Arduino IDE** used to write and upload code.
An LED (Light Emitting Diode) is a semiconductor device that emits light when current flows through it. A resistor is used in series with the LED to limit the current and protect the LED from damage.
In this experiment, the Arduino digital pin is used to supply HIGH and LOW signals to the LED, causing it to turn **ON and OFF repeatedly**, which creates the blinking effect.

Circuit Diagram

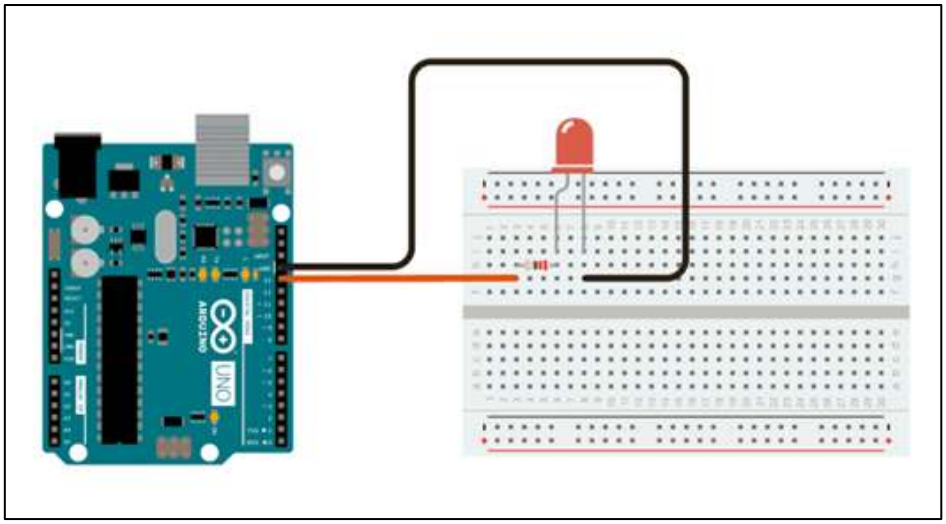

Figure 1: The Circuit Diagram of the Blinking the LED

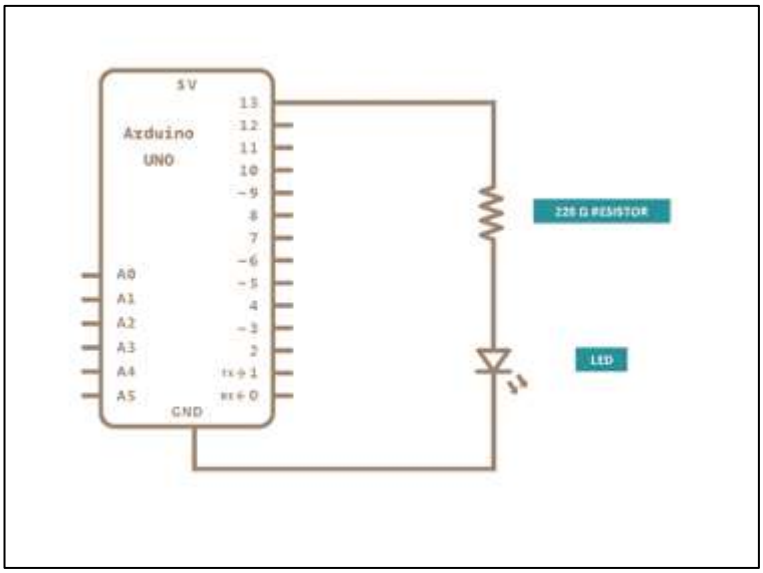

Figure 2: The Schematic Diagram of the Blinking the LED

Procedure

Circuit Setup

1. Place the LED on the breadboard.
2. Connect a 220 Ω resistor to the LED anode.
3. Connect the resistor to digital pin of the Arduino board.
4. Connect the LED cathode to GND.
5. Ensure all connections are secure.

Arduino Sketch

Open the **Arduino IDE** and type the following program.

```
int ledPin = 8;

void setup() {
  pinMode(ledPin, OUTPUT);
}

void loop() {
  digitalWrite(ledPin, HIGH);
  delay(1000);

  digitalWrite(ledPin, LOW);
  delay(1000);
}
```

Uploading the Code

1. Connect the Arduino board to the computer using the USB cable.
2. Open the Arduino IDE.
3. Select the correct Board and Port from the Tools menu.
4. Click the Compile (Verify) button to check the code for errors.
5. Click the Upload button to upload the program to the Arduino board.

Observation

After uploading the program:

- The LED connected to pin 8 turns ON for one second.
- Then it turns OFF for one second.
- This process repeats continuously, producing a blinking LED.

Conclusion

In this experiment, you learned how to control an LED using an Arduino board. By writing a simple Arduino program, the LED was blinked at regular intervals successfully. This demonstrates how Arduino digital pins can control external electronic components.